AI LAB 99

Machine Learning Track — Week 4, Module 10

Streamlit App Implementation & QA Report

Auditing the Customer Intelligence Platform Against the Module 9 Deployment Plan

Source: app.py — 724 lines, Streamlit

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Introduction

app.py implements the eight-page Streamlit “Customer Intelligence Platform” planned in Module 9: Home, Dashboard, Customer Segments, Customer Search, Customer Prediction, Visualizations, Marketing Strategy, and Download (the shipped navigation differs slightly in naming/grouping from the Module 9 plan's Home / Dataset Overview / Customer Segments / Business Dashboard / Customer Personas / Predict Customer Segment / Marketing Recommendations / About Project, but covers the same ground). This report audits each page against the risks Module 9 flagged in advance, and catalogs anything new found while reading the implementation.

Headline result: two of the three risks flagged in Module 9 materialized exactly as predicted — the dashboard displays raw z-scores with a dollar sign, and the Marketing Strategy page was shipped as static, unwired content rather than being connected to the real segment analysis. A third issue, the prediction page, was not fixed but avoided: rather than risk an incorrect prediction, the page was left as an un-implemented placeholder.

Page	Status	Finding
Home	**OK**	KPI counts (customers, features, segments) are raw counts, not scaled values — safe to display as-is.
Dashboard	**BROKEN**	“Average Income” and “Total Spend” are formatted with a $ sign directly on standardized z-score data. Income Range Filter slider is built from int(min/max) of the same z-scores.
Customer Segments	**OK**	Segment counts and percentages are computed from the categorical Customer_Segment column — not affected by scaling.
Customer Search	**PARTIAL**	Functions, but placeholder text invites searching by “Customer ID,” a column that was dropped during Module 5 feature selection and no longer exists.
Customer Prediction	**STUB**	No model is loaded and no prediction runs. The page shows a status message and a warning; the “real-time predictor” promised in the navigation label does not predict anything.
Visualizations	**PARTIAL**	Charts render, but “Income Tier Distribution” bins and “Total Product Category Revenue” are drawn from standardized values without inverse-transforming or relabeling.
Marketing Strategy	**STATIC**	Four hard-coded strategy cards (Premium, At-Risk, Budget, Digital-First) are not wired to any data — generic copy, not the Module 8 analysis output.
Download	**OK**	Full-dataset and filtered-subset CSV export both work correctly via st.download_button.

Home Page — OK

Displays total customers, total features, segment count, and a static “System Health: Active” indicator, plus the resolved dataset file path. All four values are simple counts or string constants, so the standardization issue that affects other pages does not apply here.

Dashboard Page — Confirmed Bug (Module 9 Risk #1)

This page filters customers by segment, activity level, shopping channel, and an income slider, then shows four KPI cards.

```
average_income = pd.to_numeric(filtered_df["Income"], errors="coerce").mean() total_spending =
pd.to_numeric(filtered_df["Total_spending"], errors="coerce").sum() ... st.metric("AVERAGE INCOME",
f"${average_income:,.0f}" if average_income > 0 else "N/A") st.metric("TOTAL SPEND",
f"${total_spending:,.0f}" if total_spending > 0 else "N/A")
```

This is exactly the risk Module 9 warned about: Income and Total_spending are still StandardScaler output (roughly −3 to +3), never inverse-transformed. “Average Income” will display as something like “$1” or “N/A” (since the > 0 guard will hide it whenever the filtered mean is negative), and “Total Spend” will show small or negative dollar figures for a filtered group — neither is a usable business number. The Income Range Filter slider compounds this: int(min_income) and int(max_income) truncate the z-score range to something like −1 to 3, so the filter's own label (“� Income Range Filter”) presents a 4-point range as if it were a dollar span.

Fix: load standard_scaler.pkl (already confirmed present in Module 9) and apply scaler.inverse_transform() to Income and Total_spending immediately after loading the dataframe, before any KPI, slider, or chart uses them.

Customer Segments Page — OK

Shows a segment summary table (counts and percentages) and a bar chart, both computed from the categorical Customer_Segment column. Since this column holds string labels rather than scaled numbers, this page is unaffected by the standardization issue and works as intended.

Customer Search Page — Minor Issue

A free-text search box checks whether the query string appears anywhere in any column of any row. This works mechanically, but the placeholder text reads “Search Database (Customer ID, Segment, Channel, etc.)” — the ID column was dropped from the dataset during Module 5 feature selection (it was flagged there as a pure identifier with no clustering value) and is not present in Final Dataset.csv. A user following the placeholder's suggestion to search by Customer ID will get no matches and no explanation why.

Customer Prediction Page — Not Implemented (Module 9 Risk #3, Avoided Rather Than Fixed)

```
st.info("The saved `kmeans_model.pkl` pipeline is staged for inference connection.") st.warning("� Feature
sequencing validation is required before serving live predictions to ensure exact encoder compatibility.")
```

This is the entire page. No model is loaded, no input form is shown, and no prediction is ever run. Module 9 flagged that a working prediction page would need the full preprocessing chain (missing-value handling → feature engineering → encoding → StandardScaler) applied to new input before calling kmeans_model.predict(); rather than risk shipping an incorrect prediction from a partially-applied pipeline, this page was left as a placeholder. That is a defensible short-term choice, but “� Customer

Prediction” in the navigation currently promises a feature the app does not deliver — either implement it using the Module 5 preprocessing_pipeline() function plus the saved scaler and model, or relabel the page/remove it from navigation until it's built.

Visualizations Page — Partial Issue

Two charts: an “Income Tier Distribution” histogram (10 bins of the raw Income column) and a “Total Product Category Revenue” bar chart (sum of the six Mnt* columns). Both are technically correct renders of the data they're given, but neither column has been inverse-transformed, so the bin edges and bar heights are in z-score units, not income tiers or revenue dollars. Unlike the Dashboard page, no $ sign is attached here, which makes this a lower-severity version of the same underlying issue — the chart shapes are directionally useful (relative comparison across products/customers still holds) but the axis values and the word “Revenue” in the title overstate what's actually being shown.

Marketing Strategy Page — Static Content (Module 9 Risk #2, Confirmed)

This page renders four fixed strategy cards — Premium/High-Value, At-Risk & Churning, Budget & Value Seekers, and Digital-First Shoppers — with hand-written best-practice bullet points. None of it reads from the dataframe.

This confirms the Module 9 warning about the Marketing Recommendations page. It also compounds the segment-naming problem tracked since Module 7: the underlying data has exactly two clusters (Premium Loyal Customers and Digital-First Buyers/Budget Buyers, depending which module's naming you use), but this page presents four segments as if all were real and equally data-backed. Worse, it splits the single second cluster's two historical names — “Budget Buyers” (Module 8) and “Digital-First Buyers” (Module 7/9) — into two separate cards with different strategies (“discount bundles” vs. “omni-channel push notifications”), presenting one customer segment as if it were two distinct ones with two different marketing plans. “At-Risk & Churning Customers” has no corresponding cluster at all in the current model.

Fix: reduce this page to the two segments that actually exist in the data (using one consistent name for the second cluster), and, ideally, source the bullet points from the Module 8 strategy tables (once its join bug is fixed) rather than hard-coded text, so the page updates automatically if the underlying model or segments change.

Download Page — OK

Both the full master dataset and the currently filtered subset can be exported as CSV via st.download_button. This page has no dependency on the scaling issue and works as intended. Note that both exports will still contain the standardized (not inverse-transformed) numeric columns, so anyone downloading this data for use outside the app should be told the same unit caveat documented in the Module 7 and 8 reports.

Consolidated Fix List

- **Priority 1 — Inverse-transform before display:** load standard_scaler.pkl once at startup and apply it to Income, Total_spending, and Recency immediately after load_data(), so every page downstream (Dashboard KPIs, income slider, Visualizations) automatically works in real units.
- **Priority 2 — Reconcile Marketing Strategy with real segments:** cut the page to 2 cards, matching the 2 clusters that exist, using one canonical name per segment.
- **Priority 3 — Decide on Customer Prediction:** either build it properly against the saved model + scaler + Module 5 pipeline, or remove/relabel the navigation entry so it doesn't promise a feature that isn't there.
- **Priority 4 — Fix the Customer Search placeholder:** remove the “Customer ID” suggestion, since that column no longer exists in the final dataset.
- **Priority 5 — Add a units disclaimer to Download:** a one-line note that exported numeric columns are standardized, not raw, would prevent downstream misinterpretation by anyone who pulls the CSV out of the app.

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